

BIS ASSAYING & HALLMARKING CENTRES (AHC) MANUAL

TECHNICAL INFRASTRUCTURE, FIRE ASSAY METHOD & AUDITING STANDARDS

Assaying and Hallmarking Centres (AHCs) are third-party testing facilities recognized by BIS under IS 15820 to assess the fineness of precious metals and laser-mark HUID codes.

TECHNICAL TESTING PROTOCOLS (IS 1418 & IS 2112)

- **XRF Non-Destructive Screening:** Rapid preliminary verification of alloy composition and surface purity.
- **Cupellation (Fire Assay Method) - IS 1418:** The ultimate referee chemical method for gold purity measurement with an accuracy of 1 part per 10,000.
- **Laser Marking Stations:** High-precision fiber laser marking systems to engrave BIS logo, karat purity, and 6-digit HUID code with minimum height of 0.5 mm without damaging jewellery.

MANDATORY CCTV & RECORD KEEPING

All AHCs must maintain 24/7 CCTV surveillance of the sample receiving desk, assay laboratory, and laser marking stations with minimum 90-day secure backup for BIS audit verification.